

AWS Certified CloudOps Engineer – Associate

Structured Study Guide

Practice-focused guide based on the recurring decision patterns in the 398-question practice bank.

1. VPC & Networking ■

Private EC2 → Internet

- Private subnet → NAT Gateway → Internet Gateway.

Private EC2 → S3

- Use an S3 VPC Gateway Endpoint; use a bucket policy to restrict endpoint access when required.

Security Group vs NACL

- Security Group: ENI/instance level, stateful, allow rules, no explicit deny.
- NACL: subnet level, stateless, allow and deny rules; useful for blocking a specific IP.

Route 53 Resolver

- AWS → on-premises DNS: Route 53 Resolver outbound endpoint + forwarding rule.

Exam triggers

- Private subnet needs Internet → NAT Gateway
- Private EC2 needs S3 → S3 VPC endpoint
- Block a specific IP → NACL
- On-premises DNS resolution → Route 53 Resolver

2. EC2 & EBS ■

EBS performance

- High disk queue + I/O performance problem → investigate/increase IOPS.

Moving an EC2 instance

- Create AMI → launch from AMI in the new Availability Zone → terminate the original.

3. Auto Scaling ■

Target tracking

- Maintain a metric around a target → Target Tracking.

Step scaling

- Different reaction for different levels of metric deviation → Step Scaling.

Scheduled scaling

- Predictable demand at a particular time → Scheduled Scaling.

Warm pools

- Slow startup + unpredictable scaling → Auto Scaling Warm Pool.

Scale-in protection

- Long-running job should not be terminated during scale-in → scale-in protection.

InsufficientInstanceCapacity

- Consider changing instance type and/or distributing capacity across Availability Zones.

4. S3 ■

Existing-object encryption audit

- Large numbers of existing S3 objects → S3 Inventory.

S3 lifecycle

- Less frequent access + immediate retrieval + high availability → S3 Standard-IA when appropriate.

S3 + CloudFront

- Private S3 content served through CloudFront → appropriate CloudFront origin-access mechanism.

5. IAM & Cross-Account Security ■

EC2 application access

- Use an IAM role/instance profile rather than hard-coded access keys.

Cross-account access

- Use IAM role assumption for least-privilege cross-account access.

Organizations + SCP

- Organization-wide restrictions → SCP at the appropriate OU/account level. SCPs limit maximum permissions; they do not grant permissions.

6. CloudWatch & Monitoring ■

CloudWatch

- Metrics, logs, alarms, dashboards, and operational monitoring.

CloudWatch Agent

- OS-level metrics such as memory, disk, and filesystem metrics may require the CloudWatch Agent.

Composite alarms

- Multiple alarm conditions → metric alarms combined with a Composite Alarm.

7. CloudTrail vs CloudWatch vs Config ■

Memorize

- Who made an AWS API call? → CloudTrail
- Was a resource configured correctly? → AWS Config
- CPU/memory/operational metric? → CloudWatch
- Network traffic metadata? → VPC Flow Logs
- Suspicious activity/threat detection? → GuardDuty

8. Systems Manager ■

Session Manager

- Private EC2 shell access without inbound SSH/RDP or a bastion host.

Patch Manager

- Fleet patching using baselines, maintenance windows, tags, and appropriate permissions.

Automation

- Predefined operational procedures → Systems Manager Automation runbooks.

9. CloudFormation ■

AMI IDs

- AMI IDs are Region-specific. Same template across Regions with different AMI IDs → CloudFormation Mappings.

DELETE_FAILED

- Common causes include resources still associated with security groups or S3 buckets that still contain objects.

10. Storage & Backup ■

EFS

- Individual-file recovery → AWS Backup partial restore when supported by the scenario.

RDS encryption

- Existing unencrypted RDS → snapshot → copy snapshot with encryption → restore a new RDS instance.

DynamoDB DR

- Cross-Region replication → DynamoDB Global Tables.

11. KMS ■

Imported key material

- Imported KMS key material + custom rotation schedule → new KMS key, import new material, update alias as appropriate.

12. Databases ■

Aurora

- Aurora Auto Scaling can use DatabaseConnections as the scaling metric in the relevant scenario.

Lambda + RDS

- Watch for Lambda concurrency creating excessive database connections; identify the operational control the scenario requires.

13. Cost Optimization ■

Key topics

- Savings Plans
- Reserved capacity
- Billing alarms
- Cost Explorer
- Compute Optimizer
- Organizations + consolidated billing + billing alarm: billing alerts are managed from the payer/management account in the relevant scenario.

AWS Exam Trigger Cheat Sheet

If you see...	Think...
Private EC2 → Internet	NAT Gateway
Private EC2 → S3	S3 Gateway Endpoint
Block specific IP	NACL
EC2 application needs AWS permissions	IAM Role
Cross-account access	Assume Role
Organization-wide restriction	SCP + OU
Who made API call?	CloudTrail
Configuration compliance	AWS Config
CPU/memory/disk monitoring	CloudWatch / Agent
EC2 private shell access	SSM Session Manager
Patch fleet	SSM Patch Manager
Automated operational procedure	SSM Automation
Maintain metric target	Target Tracking
Different scaling based on severity	Step Scaling
Predictable traffic	Scheduled Scaling
Slow EC2 boot during scale-out	Warm Pool
S3 existing-object inventory	S3 Inventory
S3 less frequent but immediate access	Standard-IA
S3 private behind CloudFront	OAI/OAC pattern
EC2 → different AZ	AMI → launch
EBS high I/O queue	Increase IOPS
RDS unencrypted → encrypted	Encrypted snapshot → restore
DynamoDB cross-Region DR	Global Tables
EFS individual-file recovery	AWS Backup partial restore
Imported KMS material custom rotation	New key + new material + alias
Multi-Region CloudFormation AMIs	Mappings
DNS AWS → on-prem	Resolver outbound endpoint
Long-running EC2 job shouldn't terminate	Scale-in protection
Insufficient EC2 capacity	Different instance type/AZs

How to Study the 398 Questions

- Round 1 — Learn the concepts: work through questions by topic rather than numerical order.
- Round 2 — Active recall: hide the marked answers and identify the AWS concept being tested.
- Round 3 — Exam mode: do randomized, timed practice and review every wrong answer.
- For each mistake, record: Question → Trigger → Correct AWS service → Why.

- Do not memorize answer letters; learn the underlying AWS decision pattern.

Important Source Note

This guide is based on the marked answers and recurring patterns in the uploaded practice question bank. It is a study aid for that question bank and is not an authoritative replacement for the current AWS exam guide or AWS documentation.